negative half cycle as shown in fig 10.15. In this way, current flows through load R_L always in the same direction. Hence d.c. output is obtained across R_L and this output is pulsating d.c.

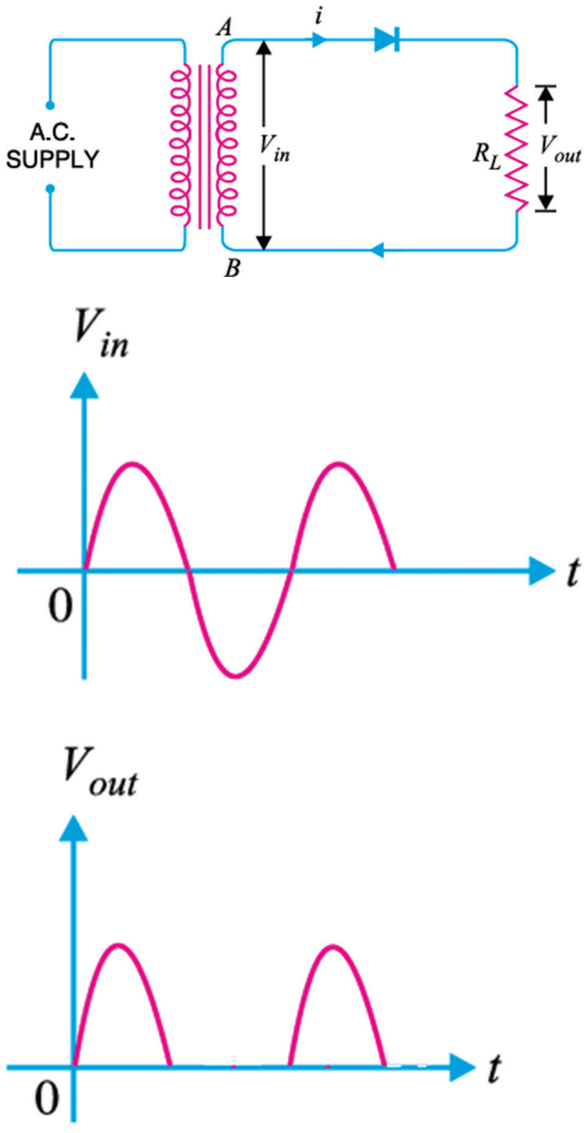


Fig 10.2 *Half wave Rectifier-wave form*

These pulsations in the output are further smoothened with the help of "Filter circuit". The peak inverse voltage of the diode should be at least equal to V_m

10.2.(ii) Full Wave Rectifier

In full wave rectification, current flows through the load is the same direction for both half-cycle of input a.c. voltage. This can be achieved with two diodes working alternately. Therefore, a full-wave rectifier utilize both half-cycle of input a.c voltage to produce the d.c.output. The following two circuits are commonly used for full wave rectification.

(i) Centre tap full wave rectifier
(ii) Full-wave bridge rectifier

(a) Centre Tap Full-Wave Rectifier

Circuit Details

The circuit employs two diodes D_1 and D_2as shown in fig 10.16. A centre tapped secondary winding AB is used with two diodes connected so that each uses one half-cycle of input a.c. voltage.

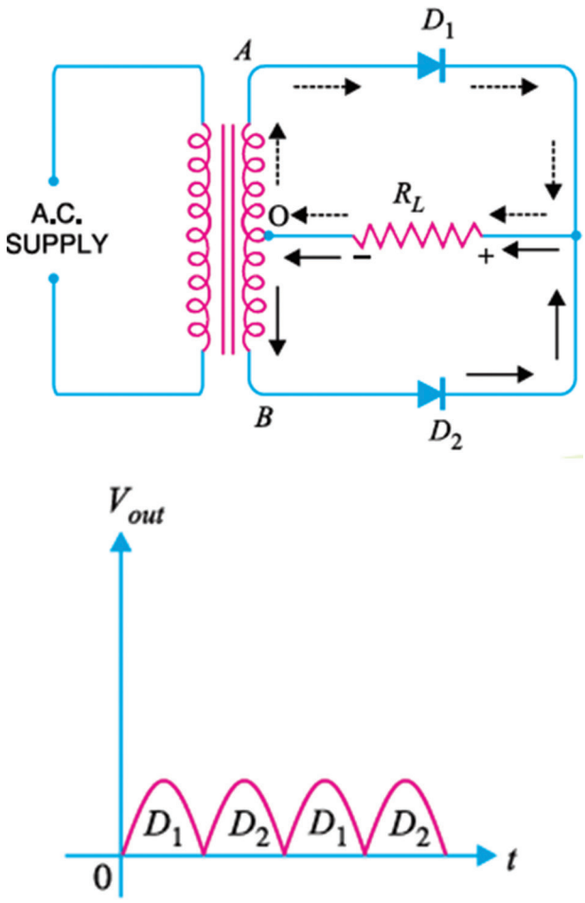


Fig 10.3 *Full wave Rectifier-Wave Form*

EMA_English_Chp10.indd 157 12/17/2019 5:54:24 PM